The Computational Organization of MoE Modularity

Learning history, property transport, and predictive cooperation

Research argument draft

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Abstract

Training sparse MoE models for selective execution raises a fundamental question: what computational organization makes their expert subsets useful? We show that selectability can improve while dependence on shared computation grows, even at similar intact quality. Cross-source interventions then separate the functions carrying a useful deployment property from the partners supporting its predictive use. Changing routed functions transfers a low-budget likelihood advantage and changes the predictive role of unchanged shared functions; both effects persist when actual expert support and weights are clamped on a 55-subject panel. Cross-context replay of realized shared messages identifies learned production–use relationships whose prediction cost largely survives local geometric correction. A second mature lineage reveals a complementary reversal: native restricted performance worsens while the transported expert bank improves restricted likelihood in a common receiver, with backbone changes opposing that improvement. Together, these results show how learning history organizes selectable functions into context-dependent computations: the boundary of a useful expert subset need not coincide with the boundary of reusable computation. The transported property improves probability quality; accuracy is reported separately.

1 What organization does modularity learn?

Sparse MoE models offer a compelling route to scalable, adaptable computation: a large bank of experts can serve different inputs through small active subsets. Training objectives can go further and make document-level expert pools useful for deployment [1]. Understanding what these objectives organize is central to the scientific promise of MoE: how tasks are divided, which functions carry transferable properties, and which computations can be reused across learning histories.

Our central finding is that these questions have different answers within the same mature systems. At similar intact quality, a model with better selective execution relies more heavily on shared computation. The routed-function package carries a low-budget predictive advantage, while its unchanged shared partner becomes more necessary in the common receiver. Realized messages retain value through their pairing with the surrounding computation. Thus, learning organizes both a selectable function bank and the relationships through which that bank supports prediction.

The paper develops this finding through a connected intervention sequence. Native histories establish the change in computational division. Source crosses identify the carrier of a useful property and the changing role of its partner. Realized-route and message interventions identify the execution paths supporting these effects. A second mature lineage demonstrates why native deployment curves alone can miss a transportable component property. The resulting account links selective deployment to the learned production and use of expert computation.

2 Objects, interventions, and evidence

We study four public mature paths from the EMO release [1]: a standard pre-annealing parent P, standard annealing A, EMO annealing B, and full-course EMO F. These architecture-matched public paths provide mature endpoints with documented training histories. Let D denote backbone, attention and readout, G the gate parameters, R the routed expert functions, and S the always-active shared functions. An R/S intervention covers the corresponding function suites in all 16 layers, defining the unit of the present comparison.

For an intact loss L, we measure selection by the loss increase under a calibration-defined expert pool, alongside fixed random pools of the same size. We measure property transport by changing a parameter source while holding a common receiver fixed. We measure a function's predictive role by its removal effect in the specified system. These measurements connect deployment behavior, parameter sources, and predictive roles.

Gate parameters held fixed do not imply realized routing held fixed: changed expert outputs can change later hidden states and gate outputs. We therefore capture actual gate logits before the common pool mask and replay them while preserving the original softmax, top-k, multiplication, and accumulation program. This clamps both support and weights. If z denotes a replayed trajectory,

the shared role in that condition is

$$\Delta_S(R, z) = L(D_P, G_P, R, 0; z) - L(D_P, G_P, R, S_P; z). \quad (1)$$

Other responses remain endogenous. The source and trajectory crosses identify controlled effects in this receiver.

Language-model measurements use 64 windows and 20 file clusters. The initial task panel contains 128 biology and mathematics questions. The route experiment adds a fixed 440-question panel: eight questions in each of the other 55 MMLU subjects, with separate validation calibration. The panels use different calibration sample counts, so between-panel magnitude differences cannot be assigned solely to task content. We report answer-token NLL, choice CE, accuracy, and output KL separately. Confidence intervals are paired input resampling intervals conditional on the models; they are not training-seed intervals.

3 Three results and their connection

3.1 Selection improves while shared dependence grows

The late- and full-course EMO endpoints B/F have intact text NLL 2.6218/2.6212 and identical task raw accuracy 39.06%. Yet F has lower selected-pool damage, while removing shared computation adds 1.8509 more text NLL damage than in B (95% CI $[1.3311, 2.3010]$). The corresponding task answer-NLL difference is 5.3112; the raw-accuracy damage difference remains unresolved. Random-pool controls show that general robustness and useful selection coexist. Selectability and cooperation increase together in these endpoints (Figure 1). On the previously specified set of 40 questions answered correctly by all four endpoints, both suites are jointly necessary in 8/10/20/24 cases for P/A/B/F. These are selected-set counts, reported alongside the full panel without population confidence claims.

3.2 A property carrier also changes the role of an unchanged partner

In the common P receiver, replacing R_P by R_B transfers a low-budget answer-NLL advantage on the original panel, whereas replacing shared functions alone does not transfer the same property. Full-quality transport fails: the original-panel full NLL increases 0.0802 while low-budget NLL decreases 0.1882. Holding S_P fixed and changing R increases its removal effect by 0.3386. The function package that carries a useful operating property and the native necessary partner must therefore be identified separately.

The larger subject panel strengthens this statement and locates its runtime implementation (Figure 2). The natural low-budget NLL advantage is 0.4207 ($[0.3221, 0.5234]$). At fixed P/B routes, the routed-function source effect remains an improvement of 0.2937/0.3824, while the unchanged shared suite's removal cost increases 0.2891/0.1840; all four intervals exclude zero. Routing still matters: the B trajectory improves R_B by a further 0.1270, and the function–trajectory interaction is -0.0887 ($[-0.1242, -0.0500]$). Routed-function responses preserve the advantage under fixed execution, and their interaction with the trajectory further shapes its size.

The choice CE advantage is 1.1386 ($[0.8431, 1.4676]$), but the raw accuracy difference is +1.14 percentage points ($[-2.27, 4.55]$). The full-regime role and low-budget property establish two parts of the organization; the same-budget shared intervention below tests their causal connection.

3.3 Realized messages retain context-dependent value

Function identity leaves production and use entangled. We therefore fix shared functions and cross their actual messages between routed-function contexts on the same input. Foreign messages have an average answer-NLL cost of 0.2605 relative to their original pairing. Correcting the consumer's clean norm and residual projection reduces this cost only to 0.2459 ($[0.2096, 0.2819]$). The improvement is 0.0146 ($[0.0050, 0.0250]$), concentrated in biology rather than mathematics; choice CE and accuracy do not show equivalent recovery (Figure 3). The tested local scalar interface does not account for most of the prediction mismatch. The residual prediction cost motivates studying the message structure and receiving computation together.

4 Native curves can conceal transportable properties

We evaluate OLMoE's public 5.033T checkpoint and its ordinary 100B annealing endpoint [10], crossing backbone, gate and expert sources under full computation and one fixed parent-calibrated pool of 16/64 experts. Its 64 routed experts and absence of an always-active shared expert extend the source–context analysis to a different architecture. All 26 predeclared conditions cover the same 55 subjects and 440 question texts, re-encoded by the OLMoE tokenizer.

After annealing, native restricted NLL damage increases by 0.1050 ($[0.0021, 0.2059]$). Yet the annealed expert bank improves restricted NLL in the common parent receiver by 0.2230 at fixed parent routes ($[0.1459, 0.3021]$), with a choice-CE improvement of 0.4876 ($[0.0844, 0.8817]$). Backbone changes oppose this property: at fixed parent gate parameters, changing only the backbone increases restricted NLL by 0.3575 with the parent bank and 0.3176 with the annealed bank (Figure 4). Native deployment

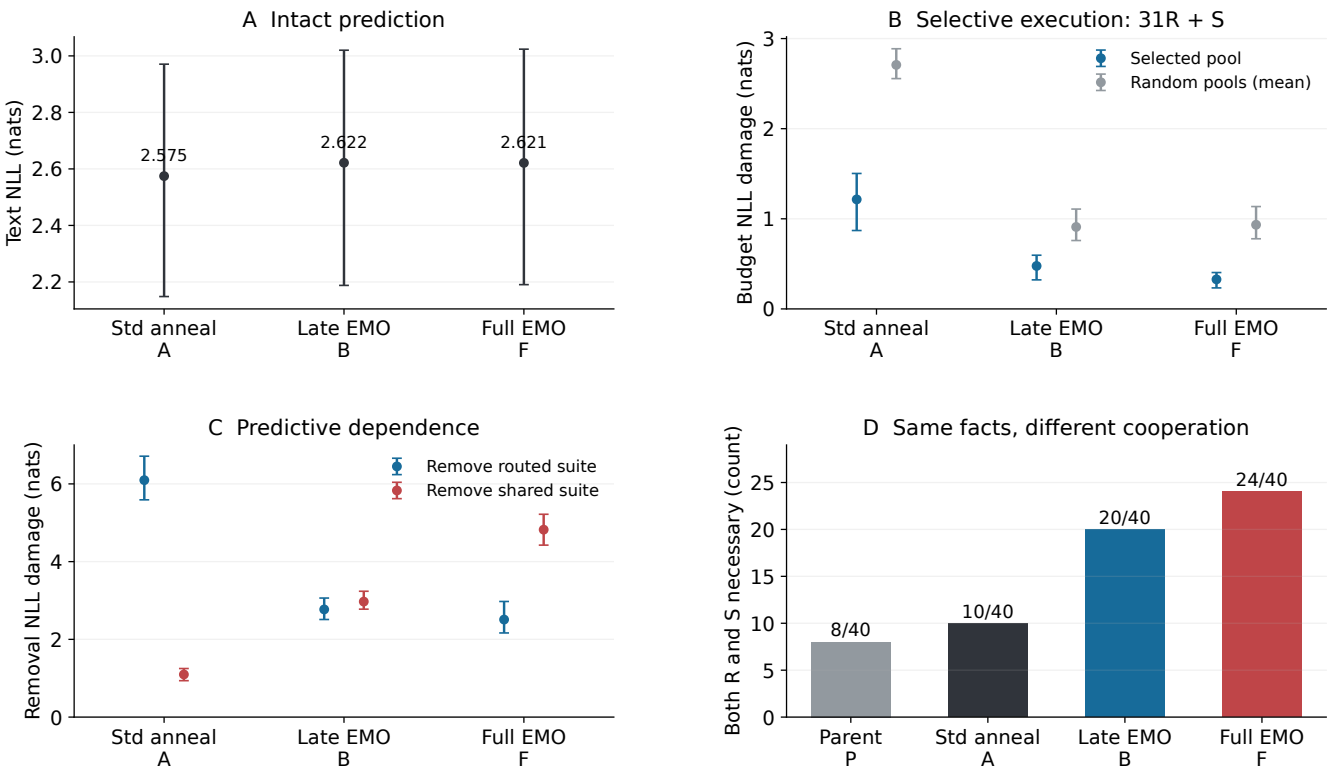


Figure 1: Computational division at similar intact quality. A–C show all three post-annealing paths, selected/random pool damage and routed/shared removal effects with original paired file-cluster intervals. D follows the same 40 questions answered correctly by all four endpoints: joint necessity counts are 8/10/20/24 for P/A/B/F. D reports selected-set counts, with no population interval. Sources: E162/E163.

curves therefore need not monotonically track a component's property in a controlled receiver.

The restricted regime is severely impaired: raw accuracy is only 22.05/21.14% compared to intact 43.64/43.18%. This intervention identifies a probability-quality property in a severely impaired restricted regime. The cross-lineage result extends the separation between a property carrier and its native operating context. Different tokenizers also preclude directly subtracting the two lineages' absolute NLL values.

The next explanatory comparison. A compact rival is output-gain calibration: expert-source changes may adjust routed amplitude while the surrounding system compensates. Prior work establishes real global-conditioning effects from massive activations and super experts [11, 12, 13]; self-selection based on internal activation is also a substantive training approach [14]. Neither a transferred likelihood advantage nor local shared-message correction by itself excludes a gain account of the routed property. The next registered comparison therefore fits one positive RMS-matching factor per layer on fixed calibration responses, without optimizing predictive loss, then evaluates parent, donor and gain-adjusted parent under identical realized routes in both lineages. For EMO, shared removal is also evaluated in the restricted regime to test whether that online partner is required for the advantage. No result from this comparison is included here.

5 Connection to MoE and interpretability research

Training and deployment. EMO trains document-level selectable expert pools [1]; interaction-aware pruning preserves cooperative structure [2]; condensation retains effective shared computation [3]. DeepSeekMoE couples fine-grained expert combinations with shared experts and evaluates routed exclusion, shared replacement, and reduced activation [9]. These works establish the value of selective and shared computation. Our intervention sequence explains how mature learning histories realize these properties through changing function sources, partner roles, and message use.

Representation and computational reuse. Spectral decoupling organizes common parameter and gradi-

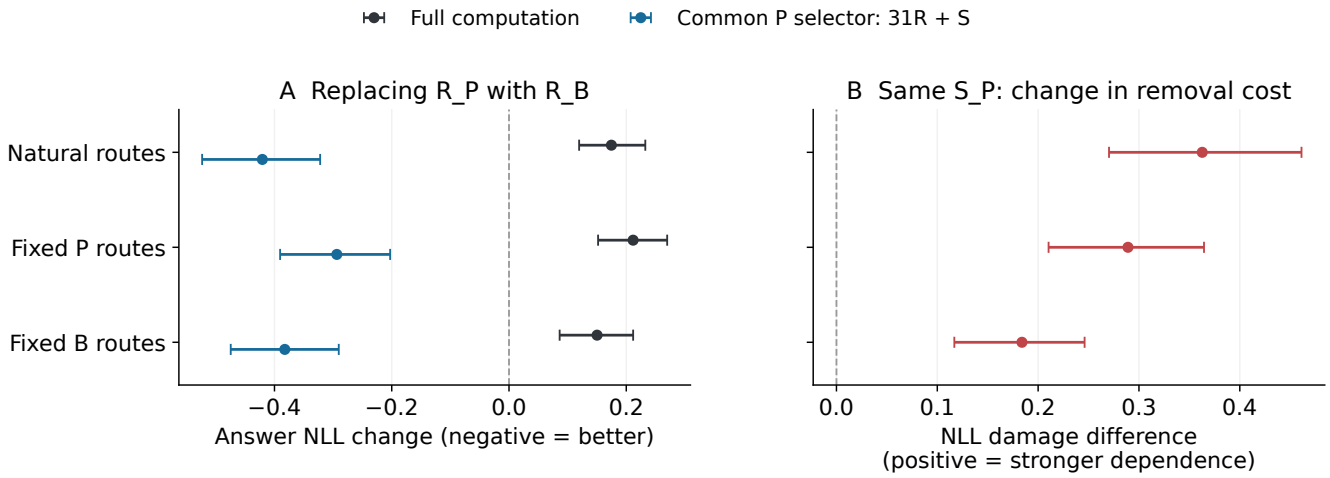


Figure 2: Two effects survive realized-route fixation on all 55 subjects. Left: changing only the routed function source improves low-budget likelihood while worsening full likelihood. Right: the same change increases the removal cost of unchanged shared functions. The actual support and weights are clamped; other response paths remain endogenous. Original subject-cluster intervals and accuracy boundary are shown. Source: E164.

ent subspaces [4]; basic–refinement attribution measures shared/routed contributions [5]; stitching tests functional-feature transport after alignment [6, 7, 8]. We connect these perspectives by following a deployment property from its source functions through realized routing to the messages used by its partners. This yields a concrete account of why a selectable component's useful property and its computational reuse can have different boundaries.

6 Evidence scope and implications

The primary objects are mature public learning paths and interventions on all-layer function suites. Input confidence intervals condition on these models and calibration data; independent initialization populations and formation trajectories remain open. The complete source grids, accuracy results, failed attempts, numerical programs and raw inputs are preserved, with JSON paths and SHA256 for every plotted estimate. Native identity replays, causal-prefix checks, source read-only checks and independent formula recomputation validate the measurement program.

The scientific implication is that the division of computation in MoE is expressed through both functions and their learned use relationships. Selective deployment, a transportable property, and reusable actual computation identify complementary parts of that organization. Understanding their relationship provides a basis for studying how computational responsibilities form, repeat across histories, and support model composition.

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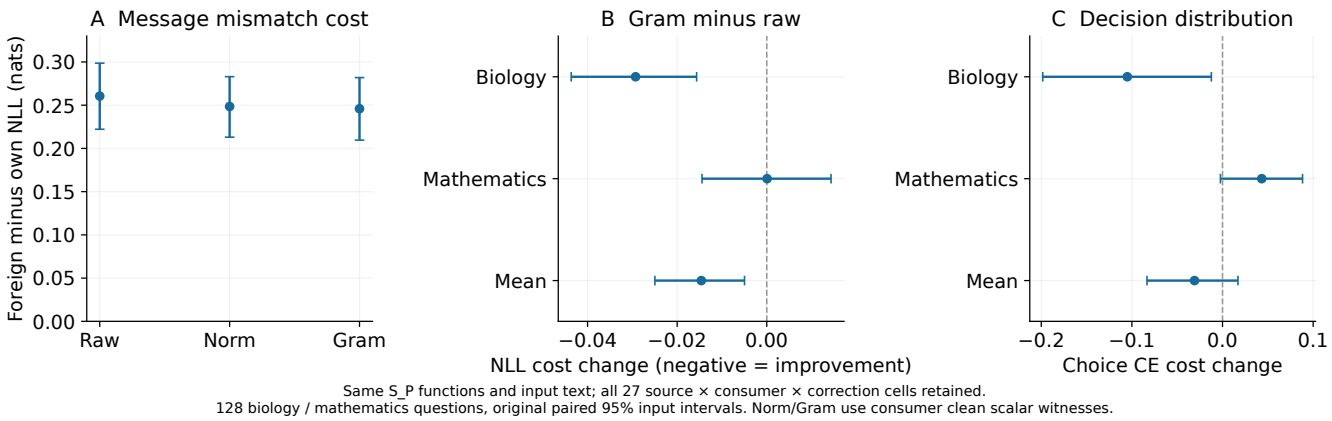


Figure 3: Actual-message reuse after local correction. All 27 source–consumer–correction cells are retained in the underlying results. Norm and Gram corrections give modest answer-NLL recovery; the subject and choice-CE panels retain the heterogeneous outcomes. Source: E163.

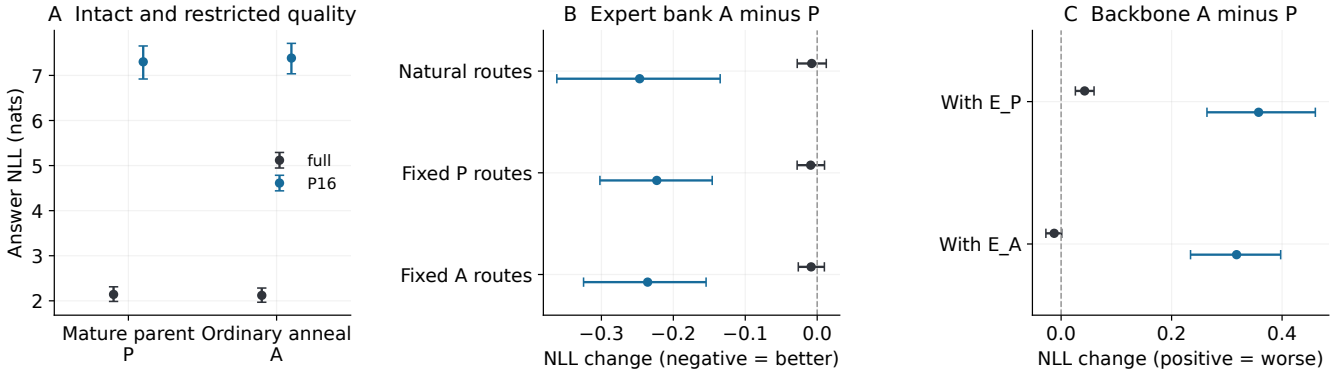


Figure 4: Native curves can conceal component properties. OLMoE expert-source changes improve restricted likelihood in the common parent receiver, including after realized-route fixation, while backbone changes oppose the improvement. Intact and restricted quality remain separate. The native restricted model is severely damaged; all source corners and accuracy boundaries are retained. Source: E165.

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